

Autonomous High-Speed Multipurpose Ground Vehicle

Mechatronics Engineer / 5-person team / Fall 2021 – Spring 2022

Overview:

The team designed and prototyped a compact, high-speed remote-controlled ground vehicle using repurposed components for future autonomous vehicle research. Key achievements included custom control integration, steering design, and safe power delivery, despite subsystem and organizational challenges. The project met core goals, was well-coordinated despite leadership gaps, and was showcased successfully at the Engineering Expo.



Prototype at NDSU Engineering Expo

Tools & Skills:

Inventor, Arduino, Stepper and DC Motor Control, Power Management, Steering Geometry Design, Team Coordination, Technical Documentation

Objective:

Design and prototype a compact, high-speed ground vehicle with remote control capability using repurposed components, intended as a platform for future research in autonomous vehicle systems and onboard sensors.

Design Highlights:

- Achieved reliable RC functionality via custom signal interfacing between hand-controller, stepper motor (steering), and drive motor controller.
- Designed and implemented steering geometry using modified off-the-shelf components.
- Developed a safe power delivery system using deep-cycle batteries with recommended safety features.
- Integrated repurposed drive components into custom systems with size and interface constraints.

My Contributions:

Led integration of control systems, power management setup, and steering mechanism. Tasks included designing steering geometry, selecting and integrating a stepper motor for steering control, managing power delivery in line with motor controller specs, and adapting control signals between the receiver and controllers. These efforts supported broader subsystem development by teammates in propulsion, suspension, and structural design. Also informally organized team meetings and deliverable submissions to support overall progress.

Reflections

The team produced a functional prototype that met performance goals within a single design cycle. Integration revealed opportunities for refinement, e.g., overbuilt body panels added weight and complexity, where thinner material would reduce cost and ease handling. A more significant issue was the independent design of steering and suspension systems, which led to dynamic interactions such as suspension compression affecting steering geometry. A more integrated approach would address these dependencies in future iterations.

Team challenges included a lack of designated leadership and unclear mentor-stakeholder boundaries. Mentors adopted a hands-off approach to encourage independence, but the team, impacted by prior COVID-related coursework disruptions, often desired clearer guidance. Without a formal spokesperson, ambiguity persisted and informal leadership remained unrecognized, slowing decisions and diluting project vision. Nonetheless, consistent meetings and shared documentation ensured effective coordination and subsystem integration.

Distributed responsibilities allowed team members to manage academic workloads while maintaining steady project progress. The prototype met key design goals and was successfully demonstrated at the NDSU Engineering Expo, drawing strong faculty and peer interest.